

Tariff Section 301 - What U.S. Companies Need To Know

Tariff Section 301

Business is shaped by the technologies we use. But technology is itself shaped by legislation. In the international business of batteries – where material sources, engineering designs, and market trends interact across borders – U.S. Section 301 tariffs are an important part of the mix, affecting any U.S. company working with Lithium-Iron Phosphate (LFP) batteries. So it may be time for companies to rethink their battery chemistry strategies- and whether a switch from LFP to Nickel Manganese Cobalt (NMC) might be beneficial.

Any such decision, however, has many parameters in play. The incoming Trump administration has announced plans for major changes to U.S. energy policy, affecting both battery suppliers' cost structures and how they'll compete with fossil sources. This will, in turn, impact EV OEMs and BESS operators. While some potential policy changes are likely to create substantial issues for businesses throughout the battery/cell value chain, others are likely to have negligible impact. Still, in a time of great transition and change, all companies can benefit from remaining agile and adaptable to shifting policy and

legislation.

Read more about these change in this blog article, revised from our Summer 2024 article of the same name.

Whatever the scenario from 2025 onwards, battery businesses will need to use the right information in the right way. And battery simulation models - like those from TWAICE - are a helpful tool in this process. In this article, you will learn why.

Section 301 tariffs for batteries are changing ...

Since 1974, Section 301 of the Trade Act has imposed tariffs and other trade barriers on imports from countries deemed to be "not playing fair" – those that enjoy cheaper labor, greater access to raw inputs, or government subsidies to export goods at prices that undercut domestic U.S. competitors. Section 301 has been used to different degrees by successive U.S. administrations, for purposes that reflect the priorities of the time.

In practice, Section 301 tariffs are most often applied to China, whose financial policies keep exports cheap – although arguably some 301 cases look more like protectionism than fairness. On the campaign trail, President-elect Trump mooted tariffs of 60% on Chinese-origin imports- falling to "an additional 10%" in more measured announcements^{[1](#)}.

However, Section 301 amendments are a negotiated

process not subject to Presidential edicts, meaning that actual increase would take many months to emerge.

The unlikely worst case: 60% tariffs on batteries

The absolute worst-case scenario for U.S. importers would be for tariffs on Chinese batteries and their raw materials to rise to 60% after 2026. This figure has been quoted by president elect Trump at campaign rallies, but is not an official part of Republican policy, so is unlikely to become reality. More likely is that federal tax credit offered by the Inflation Reduction Act (IRA) are cut short. This could result in the cost of a battery energy storage project increasing by 30-50%. Potentially, this means costs for constructors and operators more than double. Nevertheless, both these possibilities are widely thought of as extreme cases.

A more accurate picture is that IRA provisions continue much as before, and Section 301 tariffs on batteries rise no more than 10% above the Biden administration's plans, with both happening after 2027.

So there *are* meaningful changes to your cost structures ahead - but it's likely the numbers will be in the low double digits, not 60-110%. In reality, Section 301 changes may matter less to battery stakeholders than changes to the IRA timescales. But no matter the source of these changes, all battery companies must find a way to work with them.

The likely minimum: 10-20pp changes that affects business planning

In May 2024, a two-year review of Section 301 by the Biden administration ended – and it means major changes for those who use large batteries in their business models. Tariffs are doubling for some battery-related materials and nearly tripling for others, and more are still being introduced. The Trump administration is very likely to keep these. And if the President-elect's proposal to add a further 10 percentage point to Chinese imports come to pass, batteries of Chinese origin would incur import tariffs of 35%^{[2](#)}.

Given the U.S. produces relatively few battery cells domestically – and almost none of the LFP cells that are part of many American business plans – these companies are being pushed to profoundly rethink their sourcing and product strategies for everything battery-related.

There are many questions to consider. Should you stay with LFP, a relatively safe and proven option, but one that is heavily exposed to trade policy and about to become more expensive? Or should you switch product strategies to NMC chemistries, which are available domestically but may involve a potential performance trade-off and might be a less-than-ideal fit for your use case?

Political considerations: Section 301 and voting patterns

The incoming administration's tariff plans (existing plus 10% pp) would impose an additional \$450B of costs on American importers and consumers - and many of the states most impacted (those using large battery arrays and solar farms) are located in Republican-leaning districts. So, it's likely actual tariffs will be far lower than the worst-case scenario.

Adding to this the potential change to Biden's Inflation Reduction Act: if the new White House shortens the timeline on the availability of federal tax credit (currently through 2032) manufactures will have less time to get new plants up and running. Again, many of these businesses are in Republican-led districts, so it's likely to sway opinions in Congress.

TWAICE can't help with the tariffs, but we can help with your decision-making. Read on for a rundown of which factors matter and a suggested approach to modeling various scenarios featuring guest contributor Ryan Spray, Ph.D., a battery expert and principal in Exponent's polymer science and materials chemistry practice, as well as what TWAICE's Battery Analysis software brings to the party.

EVs, Li-Ion, raw materials: the tariff landscape created by Section 301

It's a complex piece of legislation, but here are the basics of Section 301 as of August 2024. Starting in 2026^{[3](#)}, all

Lithium-Ion EV batteries produced in China will be subject to 25% tariffs upon entering the U.S., up from 7.5%. The same applies to battery components, so batteries assembled in the U.S. (or any third country) from imported parts don't escape the levies.

Natural graphite – a big part of anode design in both LFP and NMC chemistries – is also getting hit for the first time, with a 25% levy⁴. Trump's stated 10% hike probably wouldn't (or couldn't) be imposed at the start of his second term, so this status quo is likely to remain into 2027 and beyond⁵. But in a sector where business must plan 5-10 years ahead, these future developments must be considered now. After all, a third of an EV's total price is its battery, and tariff changes even three years out may tip the balance between profit and loss per unit.

Note also that while the new administration is unenthusiastic about EVs and offshore wind – which use battery arrays as storage – there's another factor in play: a planned removal of limits on LNG (Liquid Natural Gas) exports from the U.S. As producers export more LNG to higher-paying non-U.S. markets, energy prices are likely to increase (by \$7.4B⁶).

On its own, this is good news for battery businesses, since it makes an already competitive energy source (battery-powered solar makes up half the local energy supply in some parts of the U.S.) even more attractive. But even with these potential advantages, the total cost for

U.S. integrators is expected to ride by 11%-16%, which could affect the economic viability of certain projects - especially those with tight margins.

There are other factors, too; mitigation by switching to alternative battery technologies isn't simply a case of costs. Can you still sign procurement agreements in the volumes you need? What is the impact of alternative choices on performance statistics for your use cases? Do multiple cells make each use case different? But some good news: with new industries hungry for clean power and CEOs increasingly focused on ESG goals - big tech companies building Ai data centers, for example - demand is going to rise for battery storage as a whole.

It's easy to see how a cost-based decision in the medium term could lead to a loss of market competitiveness in the long run. How can you answer all these questions with deadlines looming, especially if each has many iterations for you to run internally?

Let's go deeper.

Limited choices: local battery production in the U.S.

Until recently, the low prices of imported cells have made U.S.-based battery manufacturing hard to justify economically for many markets. While limited U.S. battery manufacturers have existed to supply niche industries such as aerospace, defense, and medical devices for a number of years, the options for U.S.-manufactured

Lithium-Ion cells have been even more limited. The cells and batteries produced for those niche industries have generally not been cost-competitive for mass markets and applications.

A notable exception would be that four of [Tesla](#) and Panasonic's gig factories stand on U.S. soil to supply the growth of Tesla's EV production in the U.S. However, most of this production is NMC, with LFP largely coming from Tesla's Shanghai plant. A leap in Section 301 for Chinese batteries would have strong, negative impacts for Tesla.

Domestically, there have been also changes. The U.S. excels in R&D of new battery technologies— not only new chemistries, but also new cell / electrode structures and manufacturing technologies. This R&D has led to the creation of many new battery companies with plans for large-scale manufacturing: [QuantumScape](#), [Solid Power](#), [Factorial Energy](#), [Sion Power](#), and [SES AI](#) are such examples.

While these new battery technologies still have many challenges to overcome in standing up large-scale commercial manufacturing facilities, many transportation OEMs have plans to invest in more traditional Lithium-Ion cell and battery manufacturing, though with varied timelines. With some of these timelines stretching out to 2027, it is clear that having batteries produced in the U.S. sooner rather than later will be needed to supply market

demands and take advantage of near-term incentives for domestic battery production.

The energy sector is also adapting by accelerating the setup of non-China LFP facilities to serve U.S. customers. A look into [NAATBatt Database](#) reveals a list of ten companies with projects either planned or under construction to manufacture LFP batteries in the U.S (Including LG's LFP cell factory, which is currently under construction in Arizona.)

Figure 1: List of companies with LFP battery manufacturing plans in the U.S. (from NAATBatt Database)

There's more. Those Lithium-Ion battery cells will require raw materials that need to be mined and/or supplied domestically to satisfy some of the incentive requirements. While mines that supply lithium, nickel, etc., are interested in developing more supply, the timing for developing new mines is typically longer than the timelines being planned by the aforementioned U.S. battery factories.

A prominent example of raw materials limitations is graphite for negative electrodes. The world's supply of natural graphite, while reasonably abundant among U.S. allies like Canada and Brazil, comes principally from mainland China – and is barely mined in the U.S. at all. This means a critical resource for many battery chemistries depends on supply chains stretching thousands of miles away. Additionally, Section 301

introduces a 25% levy on natural graphite coming from China - and this may increase after Jan 6, 2025.

A strategic shift for battery users across the board

Demand, of course, is growing –and the U.S. is the world's third-largest EV market. While Chinese EVs attract tariffs of 100% when exported to the U.S., China's streamlined supply chain mean they may remain competitive with U.S.-built EVs affected by the new battery tariffs. Chinese brands like BYD are already penetrating the EU market.

However, while tariff rises of a few percent can usually be absorbed in cost structures, jumps of 17.5-27,5 percentage point jumps cannot - and some large level of increase seems certain. This means any U.S. company with LFP in its business plan is heavily affected by the Section 301 changes ..

So, with the new tariffs coming into effect, companies are playing out scenarios of what remaining LFP-based versus switching to NMC could look like. Let's consider these drivers next.

Working the problem: key factors in the LFP-or-NMC decision

If you're considering a change in battery type, your first task is to gather information. But even relying on official published information carries risks. Here's a list of things

to think about.

- **Differences between real-world usage and manufacturer best-cases.** The figures in spec sheets come from labs, not customer behavior; tests are often performed under ideal conditions. It's best not to build your business strategy on specs.
- **Disparities in energy densities.** LFP cells may have lower energy densities than NMC, meaning switching to NMC may carry a weight advantage – or it may not, depending on your design and use case.
- **Endurance variations between chemistries.** LFP's greater endurance for charge-discharge cycles may be vital in calculating a vehicle's operating lifetime.
- **Effect on sustainability credentials.** NMC's use of more toxic materials may cause issues with both your green team and your customer base.
- **Externalities affecting performance.** LFPs are more prone to cold weather issues, which may dampen customer enthusiasm in certain climes.
- **LFP stockpiling as strategy.** There's evidence some market players are acquiring battery stocks before Section 301 tariffs increase, but doing so carries costs.
- **Evolving supply chains.** LFP production is ramping

up within U.S. borders but will take years to reach scale. Committing to a new technology today may miss opportunities tomorrow.

A well-informed strategy depends on accounting for multiple factors –not simply cost. What you need is **scenario modeling** based on real-world information that brings competing factors together. And that's where TWAICE comes in*.

**Dr. Ryan Spray is an independent guest contributor to TWAICE Insights, with no affiliation to TWAICE products or services.*

A solution for decision-making: TWAICE scenario modeling

The TWAICE software makes use of “physics-motivated semi-empirical aging models.” In brief, that means it combines known physical behaviors of different materials with data about how those materials perform and degrade, in different configurations and products.

Our top model goes the furthest yet: drawing on an ever-growing database of actual cells on the market to predict how your battery will change over its service lifetime. That means in addition to scenario modeling you can also use TWAICE as a third-party plausibility check ... validating a supplier data sheet against a realistic load profile.

This means you can compare different cell types –not

limited to LFP and NMC – with a high degree of confidence that the data closely models the real world, while also exploring various scenarios in countless combinations and letting you tune your business strategy against a vast range of scenario iterations (including the new realities and trade-offs Section 301 poses) in fine detail to reflect the market realities you expect. Here's a taste of the levels available:

- Our **Base Model** uses known chemistry characteristics to see how a battery changes with time and conditions. It's not cell-specific; rather it shows the general case for batteries of a given type, letting engineers model the most general cases without the need to choose a product.
- The **Customized Base Model** applies the Base Model to a specific cell product, adding data gathered from its operational specifications. Over time this model can be adjusted as real data is gathered, making it more accurate the more is known.
- The **Premium Model** is the ultimate fine-grained option, modeling not just a cell type, but a specific discrete cell, with its own external and internal conditions like configuration and temperature. This lets you model precise and detailed scenarios across all your customer use types.

Last, remember TWAICE is not a battery provider. We're an independent software developer with an interest in making our data and analytics as accurate as possible. This means TWAICE can help you plan what may be the biggest business decision you make this decade – without bias towards any particular manufacturer or chemistry, we can maximize your chance of getting it right.

Careful Analysis can turn Section 301 tariff changes into competitive advantage

Tariffs are rising from 7.5% to 25% on both complete EV batteries and their individual components. This includes new tariffs of 25% where there were none before. LFP battery supplies are largely dependent on foreign sources, and differences between battery chemistries make the decision to stay with LFP or switch to NMC technology a difficult one.

Of course, not everyone will make the best decisions about the upcoming legislation. Success belongs to those who use the right information, in the right way.

To see how TWAICE scenario models can benefit your forward planning, [contact us today](#).